

Ex: 6 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Aim:

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB.

Objective 1: MATLAB Program and Output

```
clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

mu = 0:4; % Numerology Index

bar(scs)

grid on

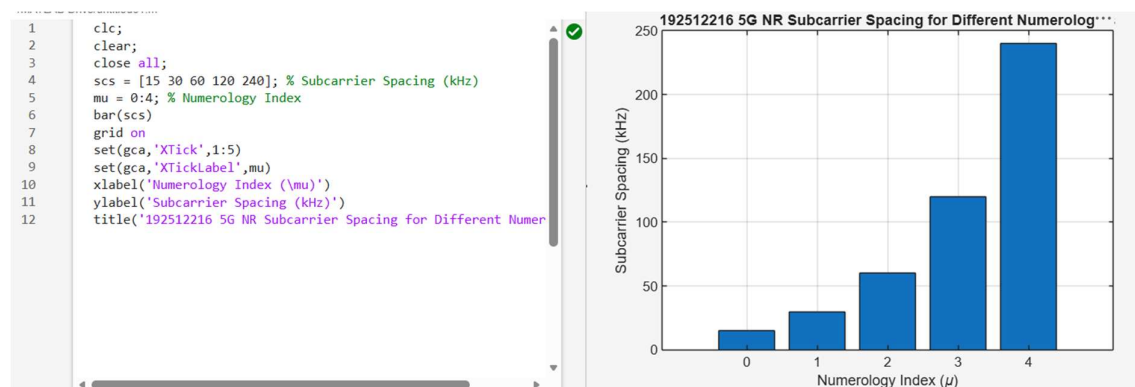
set(gca,'XTick',1:5)

set(gca,'XTickLabel',mu)

xlabel('Numerology Index (\mu)')

ylabel('Subcarrier Spacing (kHz)')

title('5G NR Subcarrier Spacing for Different Numerologies')
```



Objective 2: MATLAB Program and output

```
clear;
```

```

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)

plot(scs, slot, 'o-', 'LineWidth', 2)

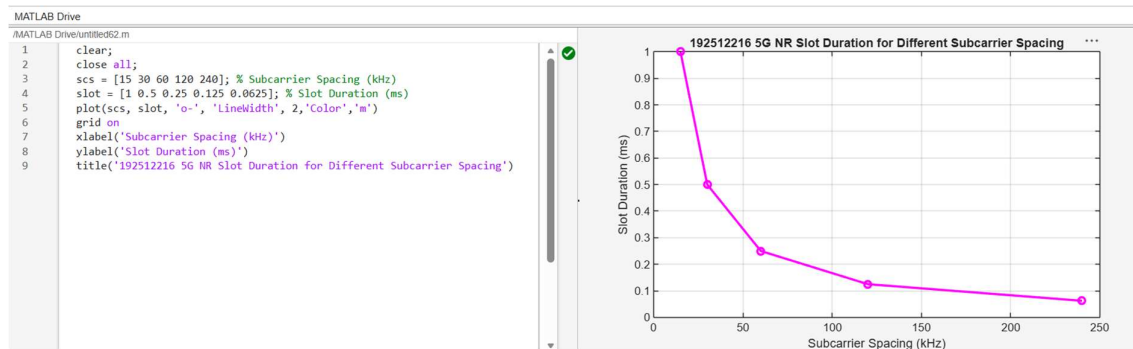
grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Slot Duration (ms)')

title('5G NR Slot Duration for Different Subcarrier Spacing')

```



Objective 3: MATLAB Program and output

```

clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

symbols = [14 14 14 14 14]; % OFDM Symbols per Slot

figure

bar(scs, symbols)

grid on

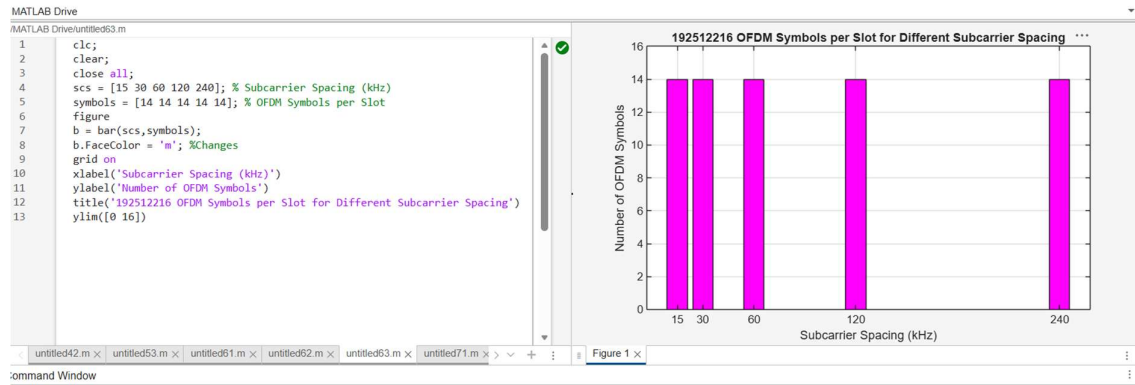
xlabel('Subcarrier Spacing (kHz)')

ylabel('Number of OFDM Symbols')

title('OFDM Symbols per Slot for Different Subcarrier Spacing')

ylim([0 16])

```



Result:

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.